1. Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms (Point 5 in the list)

This is the core ability required to solve this problem. Students need to understand the basic rules of arithmetic addition and subtraction, and apply these rules to verify a given calculation process and identify error patterns.

- **Specific Analysis:** Students first need to calculate the correct sum of the three prices: $155 + 86 + 79 = 320$ zeds. Then, they need to systematically test each option, applying the error described in each option to the correct calculation to see which result matches Olivia's incorrect answer of 248 zeds. This involves applying basic arithmetic algorithms (addition, subtraction) and logical reasoning. For example, testing option D: the correct sum of 320 zeds minus any one price, such as $320 - 72$ (assuming she subtracted 79 instead of adding it) or $320 - 79$, etc.

2. Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations (Point 7 in the list)

This task requires students to manipulate numbers and textual information, and switch between different calculation scenarios.

- **Specific Analysis:** Students need to extract the three prices (155, 86, 79) and Olivia's answer (248) from the image and text. They need to switch between various hypothetical calculation paths (options A, B, C, D) and manipulate these numbers to verify which path leads to the result 248.

3. Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation (Point 12 in the list)

Solving this problem involves a multi-step verification and reasoning process.

- **Specific Analysis:**
  1. Calculate the correct answer (320).
  2. For each multiple-choice option A-D:
     - Understand the error described.
     - Perform the corresponding mathematical calculation.
     - Compare the result with Olivia's answer of 248.
  3. Determine which option's calculation results in 248. This is a systematic process involving multiple calculations and comparisons.

4. Articulate a solution, showing and/or summarising and presenting intermediate mathematical results (Point 8 in the list)

Although multiple-choice questions typically only require selecting the final answer, the process of finding the solution requires clear logic and potential intermediate calculations.

- **Specific Analysis:** To ensure the correct choice is made, students may need to write down their calculation process on scratch paper (e.g., the result of calculating 320 with the test option D), which helps to clearly display their reasoning, even if these processes are not shown in the final submitted answer.

1. 利用基于定义、规则和形式系统的构建，包括运用熟悉的算法 (列表中的第 5 点)

这是解决此问题的核心能力。学生需要理解算术加法和减法的基本规则，并应用这些规则来验证给定的计算过程和识别错误模式。

- **具体分析：** 学生需要首先计算出三个价格的正确总和：$155 + 86 + 79 = 320$ zeds (p. 1)。然后，他们需要系统地测试每个选项，将每个选项描述的错误应用到正确的计算中，看看哪个结果匹配 Olivia 得到的错误答案 248 zeds (p. 1)。这涉及应用基本的算术算法（加法、减法）并进行逻辑推理。例如，测试选项 D：正确总和 320 zeds 减去任意一个价格，例如 $320 - 72$（假设她将 79 减去而不是加上）或 $320 - 79$ 等。

2. 操作数字、图形和统计数据及信息、代数表达式和方程式以及几何表示 (列表中的第 7 点)

该任务要求学生操作数字和文字信息，并在不同的计算场景中切换。

- **具体分析：** 学生需要从图像和文本中提取三个价格（155、86、79）和 Olivia 的答案（248）(p. 1)。他们需要在多种假设的计算路径（选项 A、B、C、D）之间切换，并操作这些数字来验证哪个路径导致了 248 这个结果。

3. 利用多步骤程序得出数学解、结论或概括 (列表中的第 12 点)

解决此问题涉及一个多步骤的验证和推理过程。

- **具体分析：**
  1. 计算正确答案（320）。
  2. 对于每个多项选择选项 A-D：
     - 理解所描述的错误。
     - 执行相应的数学计算。
     - 将结果与 Olivia 的答案 248 进行比较。
  3. 确定哪个选项的计算结果为 248。这是一个涉及多次计算和比较的系统过程。

4. 阐明解决方案，展示和/或总结并呈现中间数学结果 (列表中的第 8 点)

尽管多项选择题通常只要求选择最终答案，但找到解决方案的过程需要清晰的逻辑和潜在的中间计算。

- **具体分析：** 为了确保正确选择，学生可能需要在草稿纸上写下他们的计算过程（例如，计算 320 与测试选项 D 后的结果），这有助于清晰地展示他们的推理过程，即使这些过程未在最终提交的答案中显示。